Course: Computer Vision

Course Code: ITA0506

Slot: D

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Submitted By

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CO4_AT2_INDUSTRY PROBLEM SOLVING TASK

1. Autonomous Vehicle Vision System Selection

Given Constraints

Accuracy < 90% → System fails

Latency > 70 ms → Unacceptable

Computational resources → Moderate

Comparison

Method	Accuracy	Latency	Computation	Meets Requirements?
Viola-Jones	82%	30 ms	Low	❌
YOLO	93%	60 ms	Moderate	✅
Deep Learning	97%	90 ms	High	❌

Viola-Jones

Accuracy = 82%

Required accuracy = 90%

Since 82% < 90%, it fails.

Although its latency is only 30 ms, the accuracy is insufficient for autonomous driving.

Result: Rejected

YOLO

Accuracy = 93%

Required = 90%

93% > 90% → Pass

Latency = 60 ms

Maximum allowed = 70 ms

60 < 70 ms → Pass

Computation = Moderate

Available resources = Moderate

Pass

Result: Selected

Deep Learning

Accuracy = 97% → Pass

Latency = 90 ms

Maximum allowed = 70 ms

90 > 70 ms → Fail

It also requires high computational resources, while the vehicle supports only moderate resources.

Result: Rejected

Final Justification

YOLO is the best choice because it is the only method that satisfies all three constraints. It provides 93% accuracy, which is above the required 90%, while its 60 ms latency is below the 70 ms limit. Its moderate computational requirement also matches the available hardware.

Final Answer: YOLO

2. Smart Factory Product Inspection Selection

Given Constraints

Accuracy < 95% → System fails

Processing time > 50 ms → Production affected

Hardware → Moderate computation

Comparison

Method	Accuracy	Processing Time	Computation	Meets Requirements?
Traditional Image Analysis	91%	25 ms	Low	❌
YOLO	96%	45 ms	Moderate	✅
Deep Learning	98%	75 ms	High	❌

Traditional Image Analysis

Accuracy = 91%

Required = 95%

91% < 95% → Fail

Processing time = 25 ms → Pass

Computation = Low → Pass

Although it is fast and inexpensive, its accuracy is not sufficient to reliably detect defective products.

Result: Rejected

YOLO

Accuracy = 96%

Required = 95%

96% > 95% → Pass

Processing time = 45 ms

Maximum allowed = 50 ms

45 < 50 ms → Pass

Computation = Moderate

Available hardware = Moderate

Pass

Result: Selected

Deep Learning

Accuracy = 98% → Pass

Processing time = 75 ms

Maximum allowed = 50 ms

75 > 50 ms → Fail

Computational requirement = High

Hardware supports only moderate computation → Fail

Although it provides the highest accuracy, it is too slow and computationally expensive for this production line.

Result: Rejected

Final Justification

YOLO is the best choice because it satisfies all the required constraints. It provides 96% accuracy, which is above the 95% minimum, processes images in 45 ms, which is below the 50 ms limit, and requires moderate computation that matches the available hardware.

Final Answer: YOLO

Final Comparison

Problem	Best Method	Main Reason
Autonomous Vehicle	YOLO	93% accuracy, 60 ms latency, moderate computation
Smart Factory	YOLO	96% accuracy, 45 ms processing time, moderate computation

Overall Conclusion In both scenarios, YOLO is selected because it provides the best balance between accuracy, speed, and computational requirements while satisfying every specified constraint.